

Computer-vision pipeline for

automated mouse-behaviour scoring

What has been tried, what is in place today, and what is needed to validate the approach on the new top-down recordings.

Why a mask-feature pipeline

The objective is a scalable, label-free way to quantify scratching and grooming across cohorts, with as little manual annotation per video as possible.

SEGMENTATION

SAM3 prompt-propagation

Per-frame silhouette masks for each animal, propagated through the clip from a single prompt. Choice driven by the goal of building a classifier directly on mask geometry — keeps the pipeline label-free and avoids per-cohort retraining.

CLASSIFICATION

Hierarchical state machine

Three-level decision on mask features (velocity → posture → behaviour) with thresholds auto-calibrated from the first 300 frames. Reliable for the geometry-distinct categories (rearing, idle); ambiguous for behaviours that share a silhouette.

REVIEW LAYER

Annotation web app

React + FastAPI tool deployed at mousercv.dejandukic.dev. BORIS-grade keyboard workflow, in/out-point segments, per-mouse track lanes, undo/redo, autosave, CSV export. Built so a domain expert can review and correct labels quickly.

Front-angle recordings: where the approach hit a wall

The pipeline was first run end-to-end on the original diagonal / front-angle cohort (Cage 17082, Cage 2, DOB 160810, RJ193/RJ191). SAM3 produced clean per-frame masks, and the state machine handled rearing and idle well. The blocker emerged upstream of behaviour classification.

THE BLOCKER

Identity tracking broke down under occlusion

From the diagonal viewpoint, animals overlapped and occluded one another frequently — especially in the WT scenes with multiple mice in close contact. Reassignment errors cascaded: once two masks merged or swapped, downstream features became unreliable for the rest of the bout.

THE CONSEQUENCE

Cannot evaluate the mask-feature classifier

Without a stable per-animal mask through a full bout, the question we actually want to answer — can mask geometry alone separate scratching from grooming — cannot be tested cleanly. Tracking failure, not the classifier itself, was the limiting factor on this footage.

From raw video to reviewable behaviour timeline

Five stages, each a discrete service. Outputs of one stage become the input contract of the next, so any single layer can be swapped (e.g. classifier from rules to RF) without touching the rest.



INGEST

Video upload & storage

GCS upload · signed-URL streaming · per-cohort foldering



VISION

SAM3 prompt-propagation tracking

Per-mouse silhouette mask + centroid trajectory, propagated from a single prompt



FEATURES

Mask geometry & motion

Area · aspect ratio · convexity · velocity · oscillation spectrum



CLASSIFY

Hierarchical state machine

Velocity → posture → behaviour, with temporal smoothing into segments



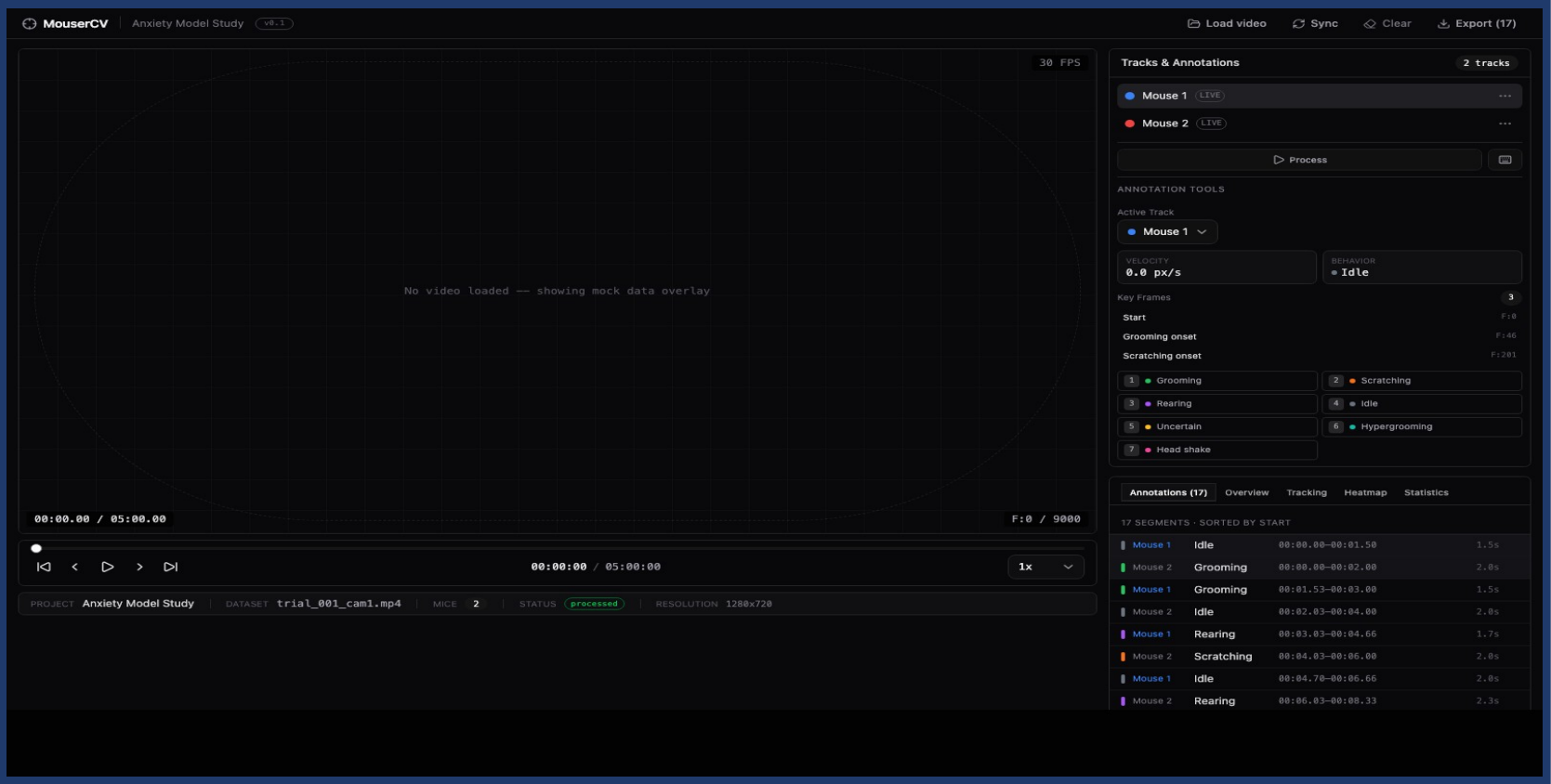
REVIEW

Annotation web app

Timeline lanes · hotkey labelling · in/out points · CSV export

Annotation tool — populated state

The review surface a domain expert lands on. Track lanes on the right, behaviour chips with hotkeys, segment list sorted by start frame, and the export hook in the top bar.



TRACK LANES

Per-mouse colour-coded lanes; identity stable across the clip from SAM3 propagation.

HOTKEYS

Seven behaviours mapped 1–7. In-point I, out-point O. Undo / redo, autosave to localStorage.

SEGMENT LIST

Default tab. Click a row to seek, hover to delete. Live count visible in the tab title.

EXPORT

CSV download with track labels, frame ranges, durations, notes — drops straight into downstream analysis.

Top-down pivot: tracking solved, questions opened

The collaboration moved to a top-down camera. Initial inspection on the available recording confirms that the tracking failure mode is gone — but the change of perspective re-opens the question of how each behaviour appears, and whether a single video is enough to draw conclusions.

RESOLVED

Animal reassignment no longer an issue

From above, mice rarely occlude one another for long. SAM3 holds identity through the clip; the multi-WT failure mode does not appear in the top-down footage.

CHANGED

Silhouette geometry is different

Postures that were distinctive from the side (rearing, hind-limb scratch) project differently from above. Existing thresholds calibrated on diagonal video do not transfer; the mask-feature set itself needs to be re-characterised.

SAMPLE LIMITATION

Only one top-down video on hand

Scratching is not prominent in this single clip, so the most behaviourally interesting category cannot yet be evaluated. A larger set of recordings — ideally enriched for scratching bouts — is the immediate gating need.

OPEN QUESTION

Mask-only discrimination — feasibility unknown

Whether mask features alone can separate scratching from grooming top-down is an open question, not a settled one. It cannot be answered without (a) more footage and (b) a small ground-truth set to calibrate against.

What unblocks the rest of the pipeline

The next steps are sequenced so that each one reduces a specific uncertainty and feeds the next. The label-free goal is preserved as the design target; whether it is achievable will be determined by the ground-truth calibration set.

01

Collect more top-down recordings

Priority on clips with clear scratching activity. Until the set is broader, no statement about top-down scratch detection can be supported by data.

02

Build a small golden-standard set

A few videos densely annotated by the domain expert — scratch, groom, hypergroom, head shake. This is what every later decision (feature choice, classifier shape, threshold tuning) will be calibrated against.

03

Validate DLC SuperAnimal keypoints top-down

Run zero-shot pose estimation (top-view-mouse model) on the new footage and check per-keypoint confidence. Keypoints are the natural fallback signal if mask features turn out to be insufficient on their own.

04

Decide classifier form against the golden set

With ground-truth in hand, test: (a) mask features alone, (b) mask + selected keypoints, (c) a learned model. Keep the simplest form that meets accuracy on the calibration set; remain label-free wherever feasible.

The annotation tool is ready to support this loop today; the bottleneck is footage and ground-truth, not infrastructure.